INTERNSHIP MINI PROJECT REPORT

SALES DATA ANALYSIS DASHBOARD

Data Cleaning • Analysis • Visualization • Business Insights

Project Detail	Information
Submitted by	PRAJWAL MALI
Internship Organization	AITECHGEN TECHNOLOGIES
Mentor Name	Not provided
Project Level	Beginner
Duration	2 Weeks
Tools	Python, Pandas, NumPy, Matplotlib, Streamlit

Prepared as part of the Sales Data Analysis Dashboard internship mini project.

1. Abstract

This project analyzes sales data using data quality checks, cleaning, exploratory analysis, visualization, and an interactive dashboard. Python and its data analysis libraries were used to calculate key performance indicators, compare product categories and regions, and identify areas for further investigation.

2. Introduction

Sales data analysis helps organizations understand revenue, profitability, product demand, and regional performance. This project demonstrates a beginner-level workflow from understanding and preparing a dataset to creating an interactive dashboard and communicating findings.

3. Project Objectives

- Understand the dataset structure and quality.
- Identify missing values, duplicate records, and date-related issues.
- Clean and prepare data for analysis.
- Calculate total sales, profit, orders, units sold, and profit margin.
- Analyze monthly sales, categories, products, and regions.
- Create charts and an interactive dashboard.
- Generate at least five meaningful business insights.

4. Tools and Technologies

- Python for programming and analysis.
- Pandas for cleaning, grouping, and aggregation.
- NumPy for numerical computing support.
- Matplotlib for static charts.
- Streamlit for the interactive dashboard.
- Visual Studio Code and PowerShell for development and execution.

5. Dataset Description

The original Excel dataset contains 10,194 rows and 21 columns. It includes order, customer, shipping, regional, product, sales, quantity, discount, and profit information. Important fields include Order ID, Order Date, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, and Profit.

6. Data Cleaning and Preparation

Quality check	Result
Original dataset	10,194 rows × 21 columns
Missing values	0
Exact duplicate rows	0
Order dates after 22 October 2026	1,019
Cleaned analysis dataset	9,175 rows

The 1,019 records with order dates after 22 October 2026 were exported separately for review and excluded from the analysis dataset. The original dataset was retained. This is a documented preparation decision; if the source is intended to contain future or planned transactions, the exclusion should be reviewed.

7. Methodology

- Data collection and understanding.
- Data quality assessment and preprocessing.
- KPI calculation and exploratory analysis.
- Category, regional, monthly, product, sub-category, and discount analysis.
- Chart creation and interactive dashboard development.
- Business insight generation and documentation.

8. Key Performance Indicators

KPI	Result
Total Sales	$2,111,931.81
Total Profit	$273,111.42
Total Orders	4,581
Units Sold	34,712
Profit Margin	12.93%
Cleaned rows	9,175

Profit margin is calculated as total profit divided by total sales, multiplied by 100.

9. Product Category Analysis

Category	Total Sales	Total Profit
Technology	$764,648.41	$135,079.78
Furniture	$681,580.73	$18,170.06
Office Supplies	$665,702.67	$119,861.58

Technology recorded the highest total sales and profit. Furniture generated substantial sales but comparatively lower profit, so its product mix, discounts, and costs merit further examination.

10. Regional Performance

Region	Sales	Profit	Orders
West	$677,194.05	$102,491.60	1,478
East	$622,585.77	$82,182.77	1,315
Central	$466,388.27	$42,735.11	1,054
South	$345,763.72	$45,701.93	734

West recorded the highest sales and profit. South had the lowest sales and order count. Central had higher sales than South but lower profit, so its product mix, costs, and discount patterns warrant investigation.

11. Monthly Sales Analysis

Month	Total Sales
December 2025	$97,502.28
September 2026	$88,064.53
September 2023	$82,670.43
November 2025	$79,411.97
November 2023	$78,826.96

December 2025 recorded the highest monthly sales in the analysis dataset.

12. Top-Selling Product

Canon imageCLASS 2200 Advanced Copier ranked first by total sales, with $51,099.85 in sales, approximately $20,159.94 in profit, 17 units sold, and 4 distinct orders.

13. Sub-Category Profitability

Sub-Category	Total Sales	Total Profit
Tables	$183,399.07	-$15,599.77
Bookcases	$108,827.77	-$3,115.69
Supplies	$43,645.85	-$1,235.62

Tables recorded the largest loss among the listed sub-categories. Pricing, discounting, costs, and product-level performance should be investigated before corrective action.

14. Discount Analysis

Discount levels of 30% and above had negative aggregate profit in this dataset. Records with a 70% discount showed total profit of approximately -$37,830.61. This association does not prove discounts alone caused losses; product mix, order characteristics, and other costs should also be examined.

15. Business Insights

1. Technology leads category performance, with $764,648.41 in sales and $135,079.78 in profit.
2. Furniture needs investigation: $681,580.73 in sales and $18,170.06 in profit.
3. West is the leading region, with $677,194.05 in sales and $102,491.60 in profit.
4. Central had $42,735.11 in profit on sales of $466,388.27.
5. Canon imageCLASS 2200 Advanced Copier leads product sales at $51,099.85.
6. Tables is the lowest-profit sub-category, with a loss of $15,599.77.
7. Discount levels of 30% and above showed negative aggregate profit.
8. December 2025 was the strongest month by sales at $97,502.28.

These are descriptive findings. Further investigation is needed before attributing results to a particular business cause.

16. Dashboard Development

The Streamlit dashboard includes KPI cards for sales, profit, orders, units sold, and profit margin; category charts; top-selling products; regional charts; monthly sales trend; sub-category profitability; dynamic insights; category, region, segment, and date filters; filtered-data download; and a filtered dataset viewer.

17. Limitations

- Results depend on the supplied dataset's accuracy and scope.
- Future-dated records were excluded and retained separately for review.
- Aggregate results can hide differences between individual products or orders.
- Relationships between discounts and profit do not establish causation.
- The analysis describes available records and is not a forecast.

18. Conclusion

The project demonstrates a beginner-level data analysis workflow using Python, Pandas, Matplotlib, and Streamlit. It includes data quality checks, cleaning, KPI calculation, category and regional comparisons, monthly trends, product analysis, visualizations, and an interactive dashboard. The findings highlight Technology category performance, the West region's leading sales, and loss-making sub-categories that warrant further review.

19. Deliverables

- Original dataset and cleaned dataset.
- Python analysis scripts and analysis outputs.
- Data quality review output and visualization charts.
- Interactive Streamlit dashboard.
- Business insights file and dashboard screenshots.
- Project report.
- GitHub repository and final presentation — still to be completed.

20. Acknowledgement

I would like to thank AITECHGEN TECHNOLOGIES and my internship mentor for providing this project opportunity. This project helped me practise data cleaning, analysis, visualization, and dashboard development using Python.